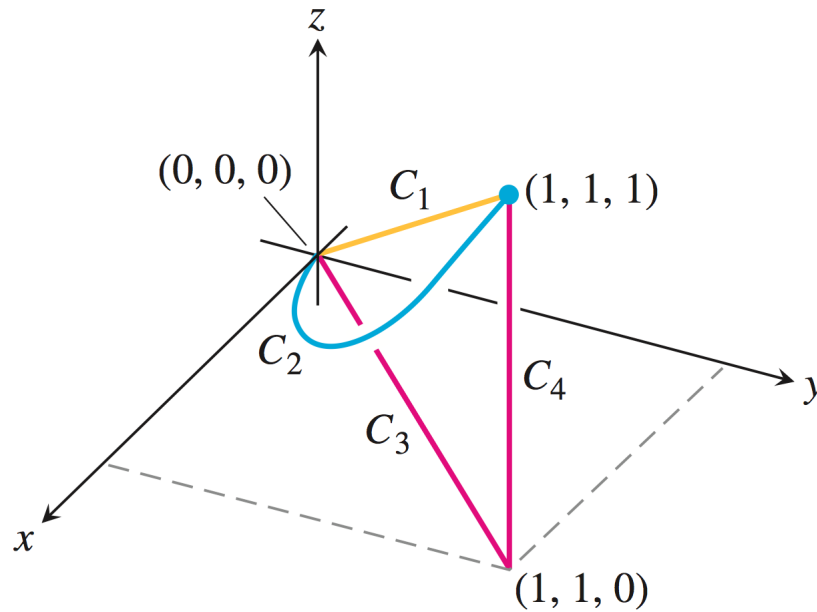


In addition to the online homework, please hand in the following problems:

Problem 1: Determine the work done in moving a mass through $\mathbf{F} = 3y\mathbf{i} + 2x\mathbf{j} + 4z\mathbf{k}$ from $(0,0,0)$ to $(1,1,1)$ along each of the following paths:

- The straight-line C_1
- The curved path C_2 (intersection of 2 parabolic cylinders: $y = x^2$ and $z = y^2$)
- The path $C_3 \cup C_4$ consisting of the line segment from $(0,0,0)$ to $(1,1,0)$ followed by the segment from $(1,1,0)$ to $(1,1,1)$



Problem 2: Find the mass of a wire that lies along the curve $\mathbf{r}(t) = (t^2 - 1)\mathbf{j} + 2t\mathbf{k}$, $0 \leq t \leq 1$ if the density is given by $\delta(x, y, z) = \frac{5}{2}z$